

Problem 10

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Problem 1 What are two substitutions that can be used to evaluate the integral

$$\int x\sqrt{x+8} \, dx$$

Explanation. Two substitutions which would work are $w = x + 8$ and $v = \sqrt{x + 8}$.

The $w = x + 8$ is the more obvious choice, so let's work through that one first. If $w = x + 8$, then $dw = dx$ and $x = w - 8$. Substituting into the original integral:

$$\begin{aligned} \int x\sqrt{x+8} \, dx &= \int (w-8)\sqrt{w} \, dw \\ &= \int (w-8)w^{1/2} \, dw \\ &= \int (w^{3/2} - 8w^{1/2}) \, dw \\ &= \frac{2}{5}w^{5/2} - 8 \cdot \frac{2}{3}w^{3/2} + C \\ &= \frac{2}{5}(x+8)^{5/2} - \frac{16}{3}(x+8)^{3/2} + C \end{aligned}$$

If $v = \sqrt{x+8}$, then

$$dv = \frac{1}{2\sqrt{x+8}}dx = \frac{1}{2v}dx \implies dx = 2v dv.$$

Also

$$v = \sqrt{x+8} \implies v^2 = x+8 \implies x = v^2 - 8.$$

Substituting these into the original integral gives:

$$\begin{aligned} \int x\sqrt{x+8} \, dx &= \int (v^2 - 8)(v)(2v) \, dv \\ &= \int (2v^4 - 16v^2) \, dv \\ &= \frac{2}{5}v^5 - \frac{16}{3}v^3 + C \\ &= \frac{2}{5}(\sqrt{x+8})^5 - \frac{16}{3}(\sqrt{x+8})^3 + C. \end{aligned}$$